ROSHAN JHA

Curriculum Vitae

School of Earth and Environmental Sciences
University of St Andrews
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PROFILE

PhD in climate science with background in hydrology and expertise in hydroclimate extremes, land–atmosphere interactions, atmospheric dynamics, climate modelling and statistics/machine learning. Current and previous research combines observations, theory, statistical and climate modelling to understand hydroclimate extremes and their risk in a changing climate.

EDUCATION

2025 PhD in Climate Science, Indian Institute of Technology (IIT) Bombay
Thesis: Heatwaves in India in a changing climate: Role of atmospheric dynamics and land–atmosphere interactions
Advisors: Dr. Arpita Mondal and Dr. Subimal Ghosh

2020 MTech in Water Resources Engineering, IIT Bombay
Thesis: Role of pre-monsoon irrigation on heatwaves in India

2017 BTech in Civil Engineering, University of Kerala, India

PROFESSIONAL EXPERIENCE

2025–Present Postdoctoral Research Fellow, University of St Andrews, UK.
Advisor: Dr. Michael P. Byrne
Project: GLOBAL-EX (A global theory for extreme land temperatures in a changing climate)

2017–2018 Design Engineer, Nepal Center for Engineering and Research, Nepal;
Projects: Storm-water management using hydrological and hydrodynamic models (HEC-HMS and HEC-RAS)

RESEARCH VISIT

2023–2024 Visiting PhD Scholar, Institute for Atmospheric Physics, Johannes Gutenberg University Mainz (JGU), Mainz, Germany
Advsior: Prof. Dr. Volkmar Wirth

PUBLICATIONS

Journal Articles

1. **Jha, R.**, & Byrne, M. P. (2026). Entrainment-based framework for understanding extratropical moist heat extremes. *Geophysical Research Letters*, 53, e2026GL122678. https://doi.org/10.1029/2026GL122678

2. **Jha, R.**, Wirth, V., Polster, C., Mondal, A., & Ghosh, S. (2025). Contrasting drivers of consecutive pre-monsoon South Asian heatwaves in 2022: Waveguide interaction and soil moisture depletion. *Journal of Geophysical Research: Atmospheres*, 130, e2024JD042376. https://doi.org/10.1029/2024JD042376

3. **Jha, R.**, Perkins-Kirkpatrick, S. E., Singh, D. et al. (2025). Extreme terrestrial heat in 2024. *Nature Reviews Earth & Environment*, 6, 234–236. https://doi.org/10.1038/s43017-025-00661-2

4. **Jha, R.**, Mondal, A., Ghosh, S., & Murtugudde, R. (2024). Northward shift of pre-monsoon zonal winds exacerbating heatwaves over India. *Geophysical Research Letters*, 51, e2024GL110486. https://doi.org/10.1029/2024GL110486

5. Lembo, V., Bordoni, S., Bevacqua, E., Domeisen, D. I., Franzke, C. L., Galfi, V. M., Garfinkel, C. I., Grams, C. M., Hockman, A., **Jha, R.**, … & Zagar, N. (2024). Dynamics, statistics, and predictability of Rossby waves, heat waves, and spatially compounding extreme events. *Bulletin of the American Meteorological Society*, 105(12), E2283–E2293. https://doi.org/10.1175/BAMS-D-24-0145.1

6. Perkins-Kirkpatrick, S., Barriopedro, D., **Jha, R.** et al. (2024). Extreme terrestrial heat in 2023. *Nature Reviews Earth & Environment*, 5, 244–246. https://doi.org/10.1038/s43017-024-00536-y

7. Zachariah, M., Arulalan, T., AchutaRao, K., Saeed, F., **Jha, R.** et al. (2023). Attribution of 2022 early-spring heatwave in India and Pakistan to climate change: lessons in assessing vulnerability and preparedness in reducing impacts. *Environmental Research: Climate*. 2, 045005. https://doi.org/10.1088/2752-5295/acf4b6

8. **Jha, R.**, Mondal, A., Devanand, A. et al. (2022). Limited influence of irrigation on pre-monsoon heat stress in the Indo-Gangetic Plain. *Nature Communications*, 13, 4275. https://doi.org/10.1038/s41467-022-31962-5

AWARDS & FELLOWSHIPS

2023–2024	Short Term Research Fellowship, DAAD
2024	Mainak Das Excellence in Publication Award, IIT Bombay
2020–2025	In-Region Scholarship Program South Asia for PhD, DAAD
2018–2020	Regional Scholarship for Master of Technology, DAAD
2018	Innovative Ideas to Post Flood Housing, Government of Kerala
2013–2017	COMPEX Scholarship for Bachelor of Technology, Government of India

CONFERENCES

2026 Role of dry-air entrainment in intensifying extratropical moist heat extremes, European Geosciences Union (EGU), 3–9 May

2024 Waveguide interaction and soil moisture depletion drive South Asian pre-monsoon heatwaves in 2022, American Geophysical Union (AGU), 9–13 December

2024 Attribution of Extreme Precipitation and Flooding in Kathmandu Valley: Synoptic Drivers, Climate Change, and Urban Impacts, 9–13 December

2023 Increasing pre-monsoon heat waves in India linked to the recent poleward shift in zonal winds, American Geophysical Union (AGU), 11–15 December

2022 Improved Estimates of Irrigation Effect on Pre-monsoon Heat Extremes in India, American Geophysical Union (AGU), 12–16 December

INVITED TALKS & SEMINARS

2025 Heatwaves in India in a changing climate, IIT Bombay, 26 August

2024 Role of waveguide interaction during 2022 heatwave in India, Goethe University, 22 January

2023 Survey of census-derived irrigation data in South Asia, Aspen Global Change Institute, 5 June

TEACHING EXPERIENCE

2026–Present Guest Lecturer, Earth and Environmental Science, University of St Andrews.

2022–2023 Graduate Teaching Assistant, Water Resources Engineering, Indian Institute of Technology Bombay.

WORKSHOPS, TRAININGS & SUMMER SCHOOLS

2026 Land–Climate Workshop, University of St Andrews, 17–19 June

2026 Climate Dynamics Workshop, Ringberg Castle, 2–4 March

2025 Scottish Climate Science Workshop, University of St Andrews, 3–5 December

2023 Climate Change Adaptation Summer School, TH Köln, 12–31 August

2023 Irrigation in the Earth System, Aspen Global Change Institute, 4–9 June

2019 GeoTraining Summer School, Goethe University, 1–27 September

DEPARTMENTAL SERVICE

2026–Present Early Career Researcher (ECR) Representative, Earth and Environmental Science, University of St Andrews

EXTRA TRAINING

2026–Present ECMWF Machine Learning for Earth Systems Modelling

2025 Google Earth Engine for Hydrological Applications

2018 ArcGIS Workshop

PROFESSIONAL MEMBERSHIPS

2026–Present European Geosciences Union (EGU)

2022–2024 American Geophysical Union (AGU)

SERVICE TO PROFESSION

Reviewer Journal of Hydrology; Journal of Climate; Geophysical Research Letters; Journal of Geophysical Research: Atmospheres; Nature Communications; International Journal of Climatology

MEDIA COVERAGE

1. Free Press Journal (2025): IIT Bombay study reveals why South Asia faced recurring extreme heatwaves in early spring 2022
2. Liang, L. (2024). Climate change amplified the effects of extreme rainfall in Nepal. *Eos*, 105. doi:10.1029/2024EO240582. Published 18 December 2024
3. Economic Times (2024): High-altitude winds shifted northwards, worsening heatwaves in the north
4. Pardikar, R. (2022). Irrigation in Indo-Gangetic Plain has little impact on heat stress. *Eos*, 103. doi:10.1029/2022EO220419. Published 30 August 2022
5. Times of India (2022): Irrigation may not be increasing summer heat stress in India: study

TEACHING AREAS

1. Eco-Hydro-Climatology
2. Global Hydrology
3. Land–Climate Dynamics
4. Hydroinformatics
5. Introduction to Climate Change
6. Introduction to Climate Modelling
7. Atmospheric Processes and Climate Change
8. Advanced Hydrological Analysis
9. Seminar-style courses built around active research questions in hydrology and climate science